



JAIPUR NATIONAL UNIVERSITY

Semester/Year End Examination Dec.2024

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science)/ B.Tech Computer Science (RPA)(In Coll.)
Semester/Year:	Semester I
Paper Code:	BTRPACO102T22/BTEPHCO103T22
Title of Paper:	Engineering Physics
Time: 3 Hours	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. Attempt any five Questions from section A and attempt any four Questions from section B.
2. Use of scientific calculator (Non programmable) is allowed.

Question Number	Question	Marks Allotted
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Section A

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| Q.1 | Write the Lorentz transformation equations. Use the Lorentz transformation to show that
$(x'^2) + (y'^2) + (z'^2) - (c^2 t'^2) = (x^2) + (y^2) + (z^2) - (c^2 t^2).$ | 4 |
| Q.2 | Explain Rayleigh's criterion of resolution. Derive an expression for the resolving power of a grating. | 4 |
| Q.3 | Write mathematical equation of time dependent and independent Schrodinger's equation. State and prove Brewster law. | 4 |
| Q.4 | Define fringe width. Find an expression for the fringe width for the bright fringes in Young's double slit experiment. | 4 |
| Q.5 | Explain what do you understand by population inversion. Derive an expression for Einstein's coefficients. | 4 |
| Q.6 | Briefly describe the construction of a optical fibre? Discuss propagation of light in single mode, multimode and graded | 4 |

index fibres.
 Q.7 Define de-Brogliewave. Write the de-Broglie relation for wavelength (λ) associated with a particle of mass m moving with velocity v . State Heisenberg uncertainty principle.

Section B

Q.1 Show that a clock moving with velocity v relative to an observer appears to him go slow by a factor of $\sqrt{1 - \frac{v^2}{c^2}}$ then at rest relative to him. Deduce Einstein's mass energy relation and discuss it.

Q.2 In Newton's ring experiment, prove that in case of reflected light (i) diameter of bright rings are proportional to the square roots of odd natural numbers and (ii) the diameters of dark rings are proportional to the square roots of natural numbers.

Q.3 Derive an expression for the intensity distribution due to Fraunhofer diffraction at a single slit. Obtain the conditions for principal maxima, minima and secondary maxima. Show that the relative intensities of successive maxima are nearly $1 : \frac{1}{22} : \frac{1}{61} : \frac{1}{121}$

Q.4 Define Laser. Draw a neat diagram of He-Ne Laser and Describe its construction and method of working. List the characteristics and applications of Laser.
 Q.5 Discuss holography and hologram. Explain the method of construction and reconstruction of hologram. What do you mean by total internal reflection, acceptance angle and numerical aperture?

Q.6 Define wave function (ψ) and give its physical significance. A particle of mass m is confined in a field free region between impenetrable walls at $x = 0$ and $x = a$. Show that stationary energy levels and wave function of

de-Broglie relation
particle of mass m
Heisenberg uncertainty

the particle are given by

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2} \quad \& \psi_n(x) = \sqrt{\frac{2}{a}} \sin \frac{n\pi}{a} x$$

relative to an
factor of $\sqrt{1 - \frac{v^2}{c^2}}$ then
mass energy

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(ii) the
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